HW1. <u>Math 515.</u>

P1. $P(A\cap B) = P(A)P(B) = 0.5\times 0.7 = 0.35$

due to independence of A and B

$$\Rightarrow P(A\cup B) = P(A)+P(B)-P(A\cap B)$$
$$= 0.5+0.7-0.35$$
$$= 0.85$$

$$P(B\setminus A) = P(B)-P(B\cap B)$$
$$= 0.7-0.35$$
$$= 0.35$$

$$P(B|A) = \frac{P(A\cap B)}{P(A)} = \frac{0.35}{0.5} = 0.7.$$

P2. (a). $\binom{5}{8}\times 0.6^5\times 0.4^3 = 0.2787$

(b). $\binom{6}{8}\times 0.6^6\times 0.4^2 + \binom{7}{8}\times 0.6^7\times 0.4$

$+\binom{1}{8}\times 0.6^8\times 0.4^0 = 0.3154.$

P3. Let

$A = \{$one of the two cards is the king of hearts$\}$

$B = \{$at least one of the two cards is a king$\}$.

$$P(A) = 1-\frac{\binom{51}{2}}{\binom{52}{2}} = \frac{2}{52} = \frac{1}{26}$$

$$P(B) = 1-\frac{\binom{48}{2}}{\binom{52}{2}} = \frac{33}{221}$$

$$P(A\cap B) = P(A) = \frac{1}{26}$$

$$\Rightarrow P(A|B) = \frac{P(A\cap B)}{P(B)} = \frac{1}{26}\times\frac{221}{33}$$
$$= \frac{221}{858}$$

P4. $f(x) = \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$E[X] = \int_{-\infty}^{+\infty} x f(x)\,dx = \int_{-\infty}^{+\infty}\frac{x}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}dx$$

$$\overset{y=\frac{x-\mu}{\sqrt{2}\sigma}}{=} \int_{-\infty}^{+\infty}\frac{\mu+\sqrt{2}\sigma y}{\sqrt{\pi}}e^{-y^2}dy.$$

$$= \mu\underbrace{\int_{-\infty}^{+\infty}\frac{1}{\sqrt{\pi}}e^{-y^2}dy}_{=1} + \frac{\sqrt{2}\sigma}{\sqrt{\pi}}\underbrace{\int_{-\infty}^{+\infty}ye^{-y^2}dy}_{=0}$$

$$= \mu$$

$$Var(X) = \int_{-\infty}^{+\infty}(x-\mu)^2 f(x)\,dx$$
$$= \int_{-\infty}^{+\infty}\frac{(x-\mu)^2}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}dx$$
$$= \frac{\sigma^2}{\sqrt{\pi}}\int_{-\infty}^{+\infty}2y^2e^{-y^2}dy$$
$$= \frac{\sigma^2}{\sqrt{\pi}}\left((-ye^{-y^2})\Big|_{y=-\infty}^{y=+\infty} + \int_{-\infty}^{+\infty}e^{-y^2}dy\right)$$
$$= \frac{\sigma^2}{\sqrt{\pi}}\cdot\sqrt{\pi} = \sigma^2$$

$$E[X^2] = Var(X)+(E[X])^2$$
$$= \sigma^2+\mu^2.$$

P5. (a) $E[X] = \int_{-\infty}^{+\infty}x f(x)\,dx$

$$= \int_{0}^{+\infty}x f(x)\,dx$$
$$= \int_0^a x f(x)\,dx + \int_a^{+\infty}x f(x)\,dx.$$
$$\geq \int_a^{+\infty}x f(x)\,dx.$$
$$\geq a\int_a^{+\infty}f(x)\,dx$$
$$= aP(X\geq a)$$

$$\Rightarrow P(X\geq a)\leq\frac{1}{a}E[X]$$

(b) Directly get by replacing X with X^2 in (a).

P6. Let X_i be the amount of the stock goes up in period i. Clearly $E[X_i]=0$.

$$Var(X_i) = 1^2\cdot\frac{1}{3} + 0^2\cdot\frac{1}{3} + (-1)^2\cdot\frac{1}{3} = \frac{2}{3}$$

$$Var(Y) = Var(X_1+X_2+X_3) = Var(X_1)+Var(X_2)+Var(X_3) = 2.$$

$$Cov(X_1,Y) = Cov(X_1, X_1+X_2+X_3) = Cov(X_1,X_1) = \frac{2}{3}$$

$$\Rightarrow Corr(X_1,Y) = \frac{Cov(X_1,Y)}{\sqrt{Var(X_1)Var(Y)}} = \frac{\frac{2}{3}}{\sqrt{\frac{4}{3}}} = \frac{1}{\sqrt{3}}$$

P7. $$|Cov(X,Y)| = \left|E[(X-E[X])(Y-E[Y])]\right| \leq \sqrt{E[(X-E[X])^2]}\sqrt{E[(Y-E[Y])^2]} = Var(X)\,Var(Y)$$

$$\Rightarrow |Corr(X,Y)| = \frac{|Cov(X,Y)|}{\sqrt{Var(X)Var(Y)}} \leq 1.$$

P8. $$E[X_1,X_2] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} x_1x_2 f_{X_1,X_2}(x_1,x_2)\,dx_1dx_2 = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} x_1x_2 f_{X_1}(x_1)f_{X_2}(x_2)\,dx_1dx_2 = \int_{-\infty}^{+\infty} x_1 f_{X_1}(x_1)\,dx_1 \int_{-\infty}^{+\infty} x_2 f_{X_2}(x_2)\,dx_2 = E[X_1]E[X_2]$$

P9. $$\mu_X(t) = E[X_t] = \sum_{k=0}^{+\infty} kP(X_t=k) = \sum_{k=0}^{+\infty} k\cdot e^{-\lambda t}\frac{(\lambda t)^k}{k!}$$

$\boxed{X_t \sim Poi(\lambda t)}$

$$= \sum_{k=1}^{+\infty} k e^{-\lambda t}\frac{(\lambda t)^k}{k!} = \sum_{k=1}^{+\infty} e^{-\lambda t}\frac{(\lambda t)^k}{(k-1)!} = \lambda t\sum_{k=1}^{+\infty} e^{-\lambda t}\frac{(\lambda t)^{k-1}}{(k-1)!} = \lambda t \underbrace{\sum_{k=0}^{+\infty} e^{-\lambda t}\frac{(\lambda t)^k}{k!}}_{=1} = \lambda t.$$

Assume $t \geq s \geq 0$, then

$$C_X(s,t) = E[X_tX_s] - E[X_t]E[X_s] = E[(X_t-X_s)X_s + X_s^2] - E[X_t]E[X_s] = E[X_t-X_s]E[X_s] + E[X_s^2] - E[X_t]E[X_s]$$

$$E[X_s^2] = \sum_{k=0}^{+\infty} k^2P(X_s=k) = \sum_{k=0}^{+\infty} k^2 e^{-\lambda s}\frac{(\lambda s)^k}{k!} = \sum_{k=1}^{+\infty} k e^{-\lambda s}\frac{(\lambda s)^k}{(k-1)!}$$

$$= \lambda s\sum_{k=1}^{+\infty} k e^{-\lambda s}\frac{(\lambda s)^{k-1}}{(k-1)!} = \lambda s\sum_{k=0}^{+\infty}(k+1)e^{-\lambda s}\frac{(\lambda s)^k}{k!} = \lambda s\left[\sum_{k=0}^{+\infty} k e^{-\lambda s}\frac{(\lambda s)^k}{k!} + \sum_{k=0}^{+\infty} e^{-\lambda s}\frac{(\lambda s)^k}{k!}\right]$$

$$= \lambda s(\lambda s+1) = \lambda^2s^2 + \lambda s$$

$$E[X_t - X_s]E[X_s] = \lambda(t-s)\lambda s = \lambda^2ts - \lambda^2s^2$$

$$E[X_t]E[X_s] = \lambda^2 st$$

Thus

$$C_X(s,t) = \lambda^2ts - \lambda^2s^2 + \lambda^2s^2 + \lambda s - \lambda^2st = \lambda s \quad \text{for } t \geq s$$

In general

$$\begin{cases} C_X(s,t) = \lambda \min(s,t) \\ \sigma_X(t) = \lambda t. \end{cases}$$